

# Himmat Stride

## Master Project & System Architecture Documentation

### 1. System Overview & Key Objectives

Himmat Stride is an Internet-of-Medical-Things (IoMT) wearable sensing platform designed for real-time biomechanical gait analysis, rehabilitation tracking, and prosthetic alignment. The system captures continuous motion data across the lower limbs using a distributed array of Inertial Measurement Unit (IMU) sensor nodes connected wirelessly to a central waist hub.

#### Primary Features

- **Wearable Multi-Node Array:** Non-intrusive strap-based attachment across key lower-limb segments.
- **Centralized Power Architecture:** A central waist-mounted battery pack powers the system, eliminating individual sensor node charging.
- **Real-Time Data Streaming:** Low-latency packet transmission over Bluetooth/Wi-Fi to mobile and PC host devices.

### 2. Hardware Architecture & Layout

#### Component Specifications

- **Sensor Nodes (6x Base + 1x Optional):** MPU6050 6-axis IMUs paired with ESP32-S3 Mini / ESP32 Mini microcontrollers.
- **Central Hub:** Waist-mounted belt enclosure housing the main controller, power bank, and wireless transceiver module.

#### Placement Strategy

- **Foot Nodes (2):** Mounted on the bridge of each foot using strap attachments.
- **Shank Nodes (2):** Front-facing placement on the lower leg segments.
- **Thigh Nodes (2):** Front-facing placement on the upper leg segments.
- **Pelvis / CoG Node (1, Optional):** Center of Gravity placement for trunk motion and balance tracking.

#### Physical Wiring (MPU6050 ↔ ESP32 Mini)

| MPU6050 Pin | ESP32 Connection | Description             |
|-------------|------------------|-------------------------|
| VCC         | 3.3V / 5V        | Operating power supply  |
| GND         | GND              | Common ground reference |
| SDA         | SDA Pin          | I2C Serial Data line    |

| MPU6050 Pin | ESP32 Connection | Description           |
|-------------|------------------|-----------------------|
| SCL         | SCL Pin          | I2C Serial Clock line |

### 3. Firmware & Data Pipeline

#### Firmware Initialization

The ESP32 Mini initializes I2C communication (Wire library) and configures the MPU6050 range parameters (excluding temperature readings to optimize payload size):

- Accelerometer Range:  $\pm 8g$
- Gyroscope Range:  $\pm 500^\circ/s$
- Filter Bandwidth: 21 Hz low-pass filtering
- Sampling Rate: 50 Hz (20 ms interval)

#### Telemetry Packet Schema

Data packets are output in CSV format for local logging and wireless transmission:

Payload Format: [node\_id, timestamp\_ms, ax, ay, az, gx, gy, gz]

### 4. Software & Cloud Ecosystem

[IMU Nodes] → I2C → [Node ESP32] → BLE/Wi-Fi → [Central Hub] → USB/BLE → [Mobile App / PC] → Cloud API → [SaaS Dashboard]

#### Module Responsibilities

- **Device Level:** Raw motion data acquisition and packet assembly.
- **Mobile App:** Bluetooth connectivity, session start/stop, local CSV session logging, and cloud synchronization.
- **SaaS Web Dashboard:** Patient account management, automated gait parameter calculations, progress trend graphs, and PDF report generation.

#### Extracted Gait Outputs

- **Spatiotemporal Parameters:** Step Length, Step Time, Cadence, Stance Phase %, Swing Phase %.
- **Kinematic Parameters:** Joint Angles (Hip, Knee, Ankle), Range of Motion (ROM), Angular Velocity, Acceleration.
- **Symmetry Analysis:** Left vs. Right leg kinematic comparison and phase alignment.

### 5. Team Structure & Core Responsibilities

| Name                | Role                        | Core Responsibilities                                                                                                   |
|---------------------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Muhammad Anees Ijaz | Founder & CEO               | Product direction, MEMS sensor strategy, incubator reporting, legal/financial structure, and business development.      |
| Zahra Gull          | Clinical Co-Founder & COO   | Clinical gait output metrics, prosthetics/orthotics rehabilitation workflows, operational management, and user testing. |
| Momina Rizwan       | Technical Co-Founder & CTO  | Technical leadership, system architecture planning, hardware/software integration, and technical documentation.         |
| Unzila Zeb          | Operations & Technical Lead | Embedded systems support, sensor-controller communications, task tracking, and administrative workflows.                |